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PerSent - A Python Library for Persian Sentiment and Emotion Analysis

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Software

- Review
- Repository
- Archive

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Summary

PerSent is an open-source Python library designed for sentiment and emotion analysis in Persian texts.
By combining machine learning models and lexical resources, it provides researchers and developers with an easy-to-use toolkit for:

- Sentiment classification of user opinions and product reviews (`recommended`, `not_recommended`, `no_idea`).
- Emotion detection across seven categories: happiness, sadness, anger, fear, disgust, surprise, and calmness.
- Batch analysis of large datasets with CSV input/output.
- Model training and persistence for customized use.
- provide pre-trained model

The library is available on PyPI under the MIT License and is intended to fill the gap in public, reusable resources for **Persian Natural Language Processing** (NLP).

Statement of need

While sentiment analysis tools are abundant in English and other widely spoken languages, resources for **Persian** remain scarce, fragmented, and often proprietary.
This lack hinders both academic research and industrial applications in NLP for Persian text.

PerSent addresses this gap by offering:

1. **Unified functionality** — a single Python interface for both sentiment and emotion analysis in Persian.
2. **Flexibility** — supports both pre-trained models (~80% baseline accuracy) and custom dataset training.
3. **Batch and interactive modes** — suitable for both large-scale corpus analysis and real-time processing.
4. **Ease of integration** — lightweight dependencies and clear API make it easy to embed into websites, chatbots, or research workflows.

By publishing PerSent as an open-source, PyPI-installable package, we enable NLP researchers, computational linguists, and developers in Persian-speaking contexts to prototype and deploy faster.

Functionality

The library provides two main analyzers:

- **CommentAnalyzer** — trains and applies models for recommendation-based sentiment.
- **SentimentAnalyzer** — lexicon-based or trained models for multi-label emotion detection.

Example usage:

```
from PerSent import CommentAnalyzer

analyzer = CommentAnalyzer()
analyzer.loadModel()
print(analyzer.analyzeText("□□□□ □□□□ □□□□□"))  # recommended
```

Batch CSV analysis:

```
analyzer.analyzeCSV("comments.csv", "results.csv", "summary.csv")
```

Emotion analysis:

```
from PerSent import SentimentAnalyzer
analyzer = SentimentAnalyzer()
analyzer.loadModel()
print(analyzer.analyzeText("□□□□ □□□□ □□ □□□□□□□."))  # The output includes percentages o
```


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References

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